Training Continuous Chain of Thought Models: A Tale of Two Regimes

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ContinuousCoT approaches train models to reason in few dense vectors instead of long token sequences **making CoT efficient!**

Current methods (COCONUT, CODI, Sim-CoT) **indirectly** supervise dense vectors to match CoT traces, requiring **slow** recurrent forward passes during training.

RQ1: How can we efficiently train ContinuousCoT models?

C-MTP; Direct supervision via Multi-Token Prediction.
1 forward pass / training step.
No recurrence.

RQ2: Do ContinuousCoT approaches — direct or indirect — perform well across diverse evaluation settings?

✓ **Simple tasks** [GSM8k-Aug]
∼**80%** of CoT accuracy

× **Realistic tasks** [MATH]
∼**30%** of CoT accuracy
→ both regimes collapse

Simple benchmarks overstate ContinuousCoT's effectiveness for realistic settings.

Simple: Easy problems, short traces

Structured (∼25 tokens on avg)

≪18*12=216≫ ≪216-6*12=144≫ ≪144/12=12≫

Semi-Natural (∼62 tokens on avg)

Bella initially had 18*12 = 216 chocolates.
After giving away, she has 216 - 6*12 = 144 chocolates left. Thus, 144/12 = 12 dozens.

Realistic: Hard problems, long traces

Realistic (∼350 tokens on avg)

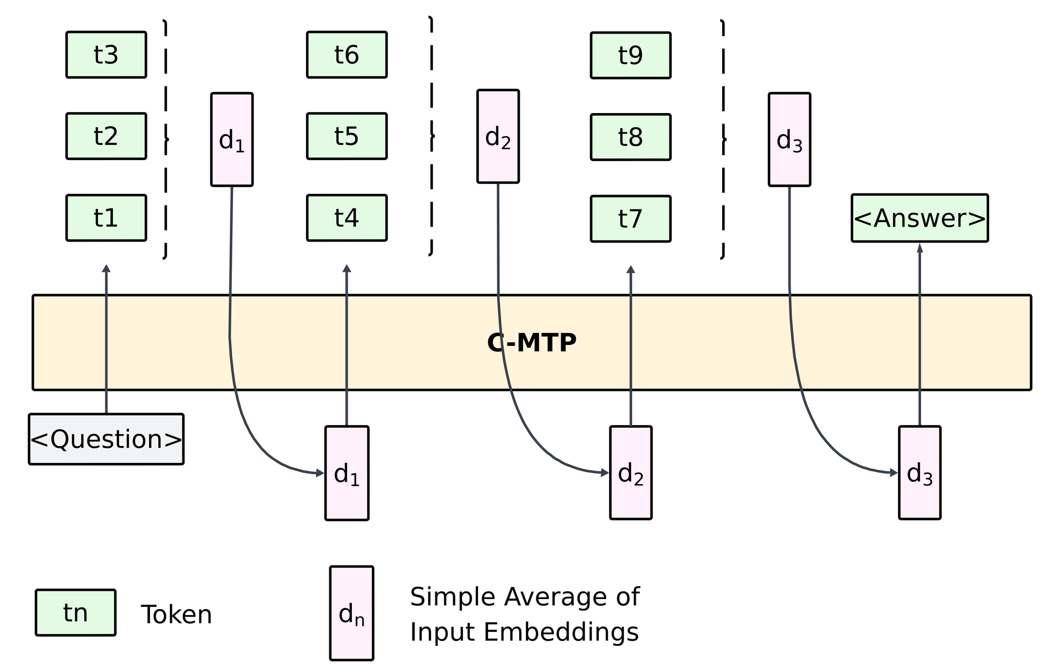
Step 1: Multiply both sides by $6x$: $5x^2+6x=18$. *Step 2:* Rearrange: $5x^2+6x-18=0$. *Step 3:* Quadratic formula: $x=\frac{-6\pm\sqrt{36+360}}{10}$. *Step 4:* Simplify: $x=\frac{-6\pm\sqrt{396}}{10}$. *Step 5:* $x=\frac{-6\pm6\sqrt{11}}{10}$. *Step 6:* $x=\frac{-3\pm3\sqrt{11}}{5}$. *Step 7:* Identify $a=-3$, $b=3$, $c=11$, $d=5$. *Step 8:* $\frac{acd}{b}=\frac{(-3)(11)(5)}{3}=\boxed{-55}$

C-MTP: Our Method

Key Idea: Latent := Aggregate (Token Embeddings)

Input: Aggregate of Token Embeddings
Output: Multi-Token Prediction

Continuous CoT *that is* **Aligned with Pretraining**



	INDIRECT Supervision Eg. CODI Coconut, Sim-CoT	**DIRECT Supervision** Eg. C-MTP CoLaR
Input	Final Layer Activation	Aggregate Token Embedding
Output	Continuous Vector	Multiple Tokens
Efficiency	Slow (>1 fwd pass)	Fast (1 fwd pass)
Reasoning Length	Fixed	Variable
Method	Recurrence	Teacher Forcing

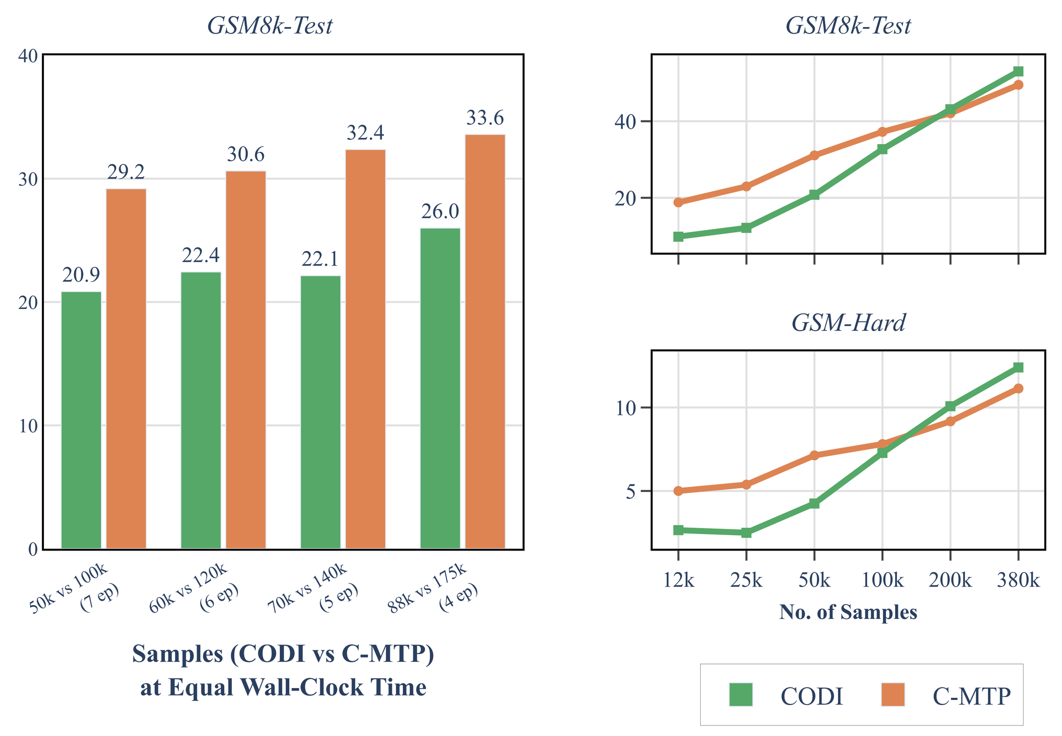
Takeaways: Semi-Natural/Realistic

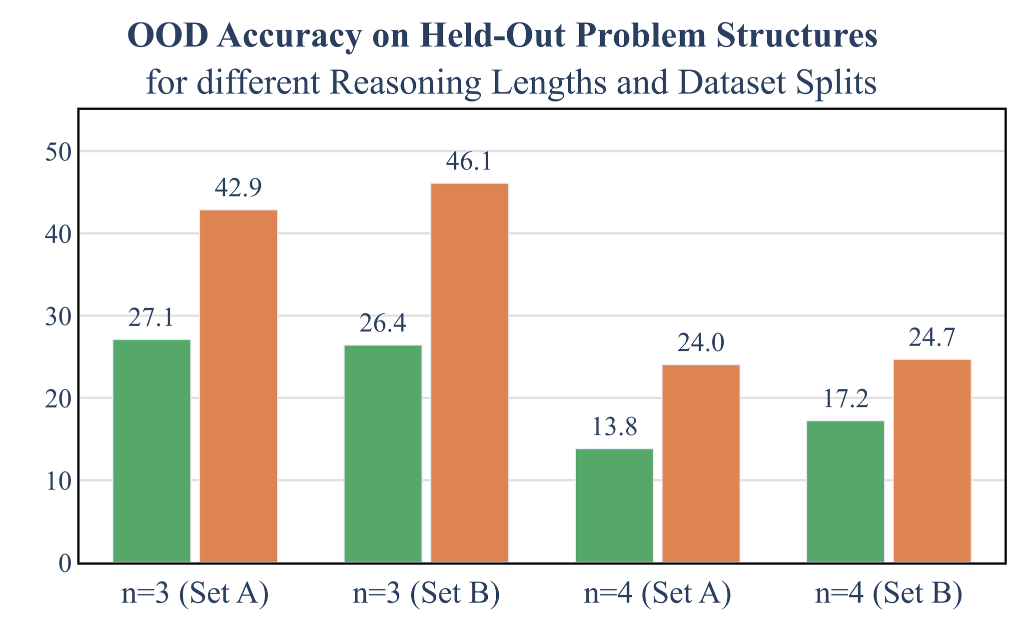
✓ **Indirect:** Fixed latent budget → compresses filler well ☺ semi-natural. ∼350 tokens into few latents → catastrophic loss. Longer recurrences: slower, unstable ☹ realistic

✓ **Direct:** Teacher forcing → train/test mismatch. Arithmetic errors at MTP span boundaries (wrong digits, dropped signs) → cascade through all subsequent predictions. ☹ both

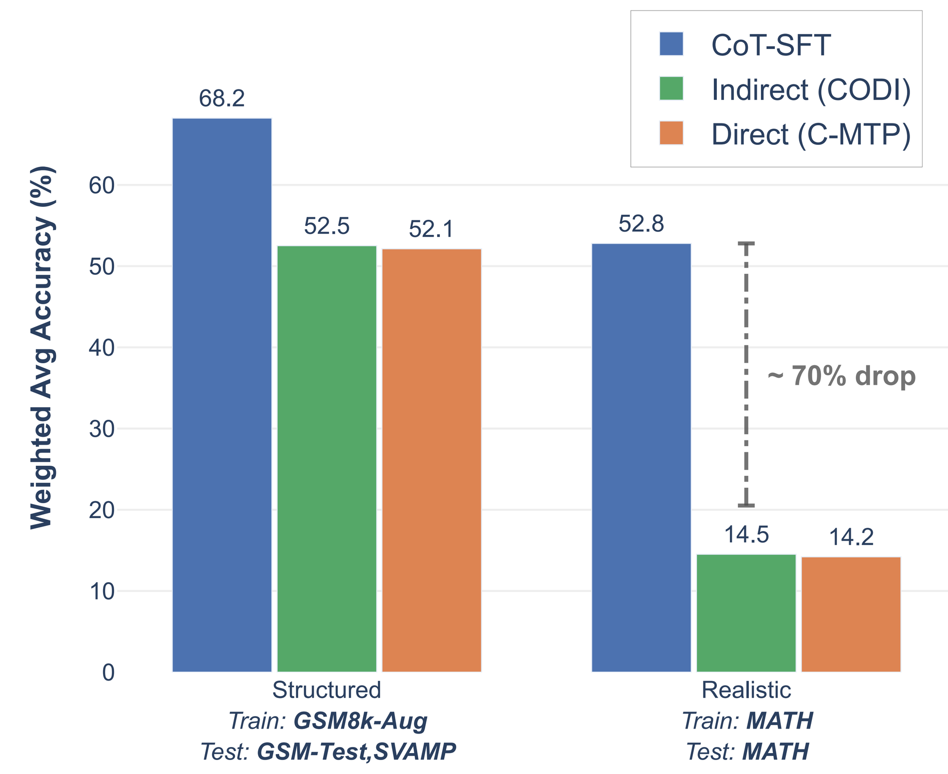
Takeaways: Simple Tasks

✓ C-MTP outperforms existing direct methods & matches indirect approaches while being 2x faster, sample efficient, and generalizing better.





Accuracy on Qwen2.5-1.5B-Instruct



Both backbones show the same trend: competitive on simple, collapse on realistic.

Neither regime alone suffices; hybrid approaches combining direct (discrete) and indirect (continuous) reasoning are a natural next step.

Accuracy vs. Reasoning Length for all Methods (Llama-3.2-1B-Instruct)